

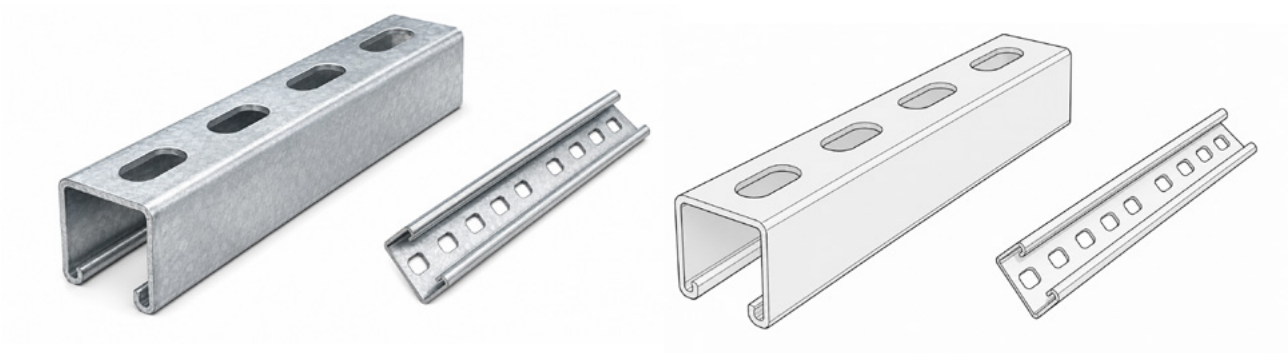


C-CHANNEL 21×41×1.0 mm



High-performance pre-galvanized and hot-dip galvanized C-Channels designed for structural framing, suspension, and support systems, manufactured to meet international standards (ASTM/EN/DIN).

### Product Specifications:



#### **K-Five C-Channel**

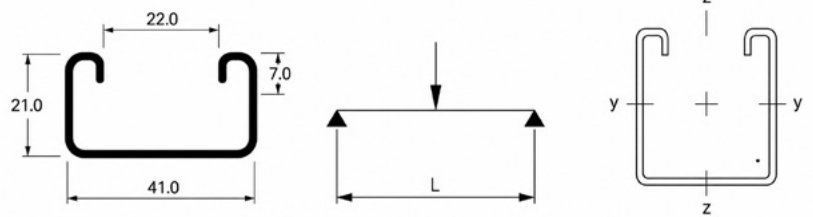
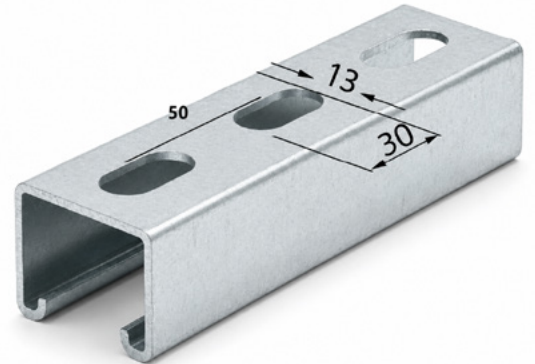
Material Finish: Pre-Galvanized /HDG

Standard Length :3 Mtr

Custom lengths available upon requests (Max. Length 6 Mtr)

## Structural Properties:

| Property                | Symbol            | Unit            | Value  |
|-------------------------|-------------------|-----------------|--------|
| Cross Section Area      | A                 | cm <sup>2</sup> | 0.81   |
| Moment of Inertia       | I <sub>y</sub>    | cm <sup>4</sup> | 0.3568 |
| Moment of Inertia       | I <sub>z</sub>    | cm <sup>4</sup> | 2.1746 |
| Section Modulus         | S <sub>y</sub>    | cm <sup>3</sup> | 0.836  |
| Torsional Constant      | IT                | cm <sup>4</sup> | 0.0027 |
| Plastic Moment Capacity | M <sub>pl,y</sub> | kNm             | 0.114  |
| Self-Weight             | G                 | kg/m            | 0.793  |



## Load Tables:

The following values represent allowable pointed & uniform loads for simply supported beams under uniform load with deflection limits at L/240 According to AISC (Steel Design Manual) with STAAD Pro. reports

| Span (cm) | Allowable pointed Load (N) | Allowable uniform Load (N/mm) | Deflection L/240 (mm) |
|-----------|----------------------------|-------------------------------|-----------------------|
| 50        | 575                        | 1.15                          | 2.08                  |
| 100       | 141                        | 0.141                         | 4.14                  |
| 150       | 59                         | 0.039                         | 6.25                  |
| 200       | 36                         | 0.018                         | 8.33                  |
| 300       | 16                         | 0.0053                        | 12.5                  |





## Design Notes & Installation

- Installation spacing should not exceed 1.2 m for typical loads.
- Recommended for use with K-Five brackets and anchor systems.
- For heavy or dynamic loads, consult K-Five technical support for custom design.
- Calculations based on allowable stress design per EN 1993-1-1.

## Quality & Standards

Manufactured under ISO 9001:2015 quality management system.

Material traceability and test certificates available upon request.

Galvanizing per EN 10346 with coating options Z140 to Z275.